MDM

Subject :- ME24052 – Introduction to Robotics & Automation

Topic:- Fundamentals of Robotics



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Unit-1: Fundamentals of Robotics	**[07 Hrs.]**
Historical development of Robotics, Definitions of Industrial Robot, Type and Classification of Robots, Asimov's laws of robotics, Robot configurations, Robot Components, Robot Degrees of Freedom, Work volume and work envelope, Robot Joints and symbols, Robot Coordinates, Robot Reference Frames, Resolution, accuracy and precision of Robot.	

Definition

- Official definition of robot was given by **Robot Industry Association (RIA)**, formerly known as Robot Institute of America.

"A robot is a reprogrammable, multifunctional manipulator designed to move material, parts, tools or specialized devices through variable programmed motions for the performance of a variety of tasks"

HISTORY

- The term "robot" was first used in 1920 in a play called "R.U.R." Or "Rossum's universal robots" by the Czech writer Karel Capek.

- The word “Robot” comes from the word “Robota”, meaning, in Czech”, forced labour,drudgery.

Many improvements

- 1959 – first commercial robot introduced by planet corporation controlled by switches
- 1960 – first unimate robot introduced for manipulator control
- 1966 – Trallfa, built and installed spray painting robot
- 1968 – mobile robot named “shakey”
- 1971 – stanford arm, a small electrically powered robot arm

- 1973 – first computer type robot programming language developed. (AL , WAVE)
- 1974 – invention of all electric drive robot
- Followed by industrial implementations for manufacturing works
- 1979 – development of SCARA type robot
- 1982 – IBM introduced Robots for assembly using robotic arm
- 1990's – invention of walking robots and rehabilitation robots, space robots, defense applications
- 2000's – Micro and Nano robots using smart materials, underwater and ariel vehicles

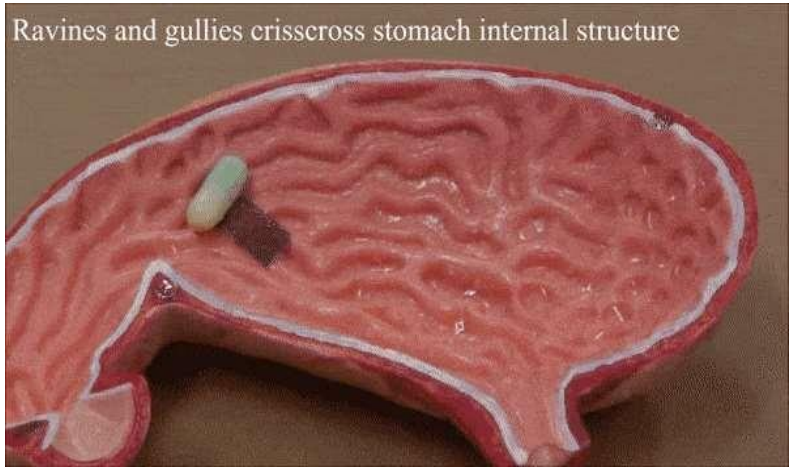
Future robots - Personal Robots

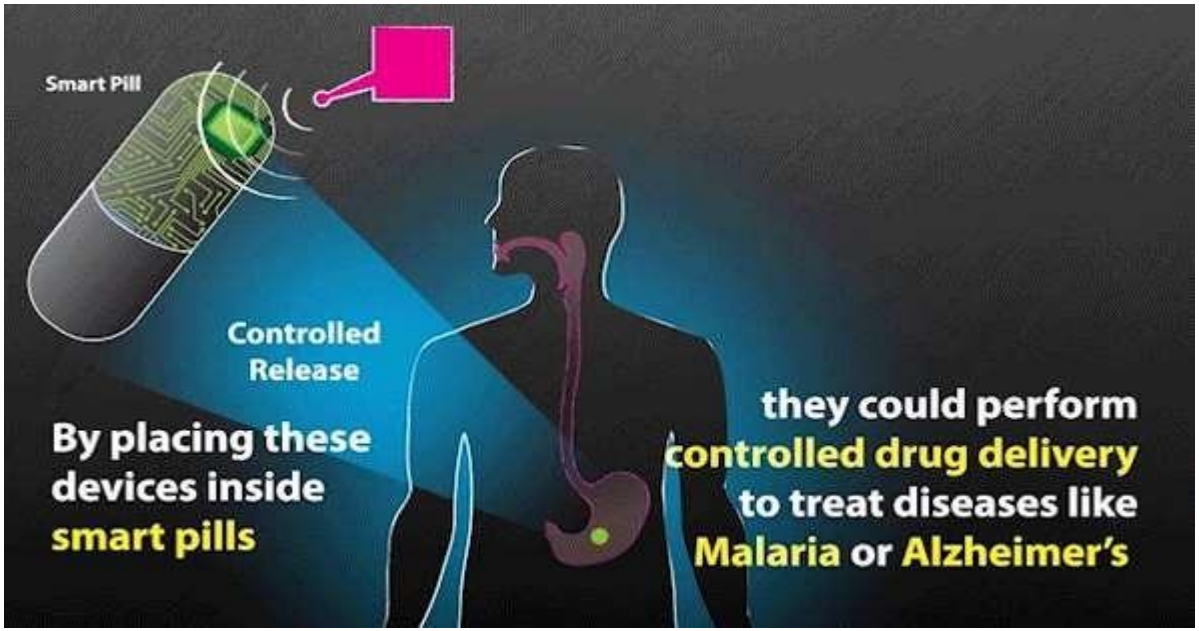




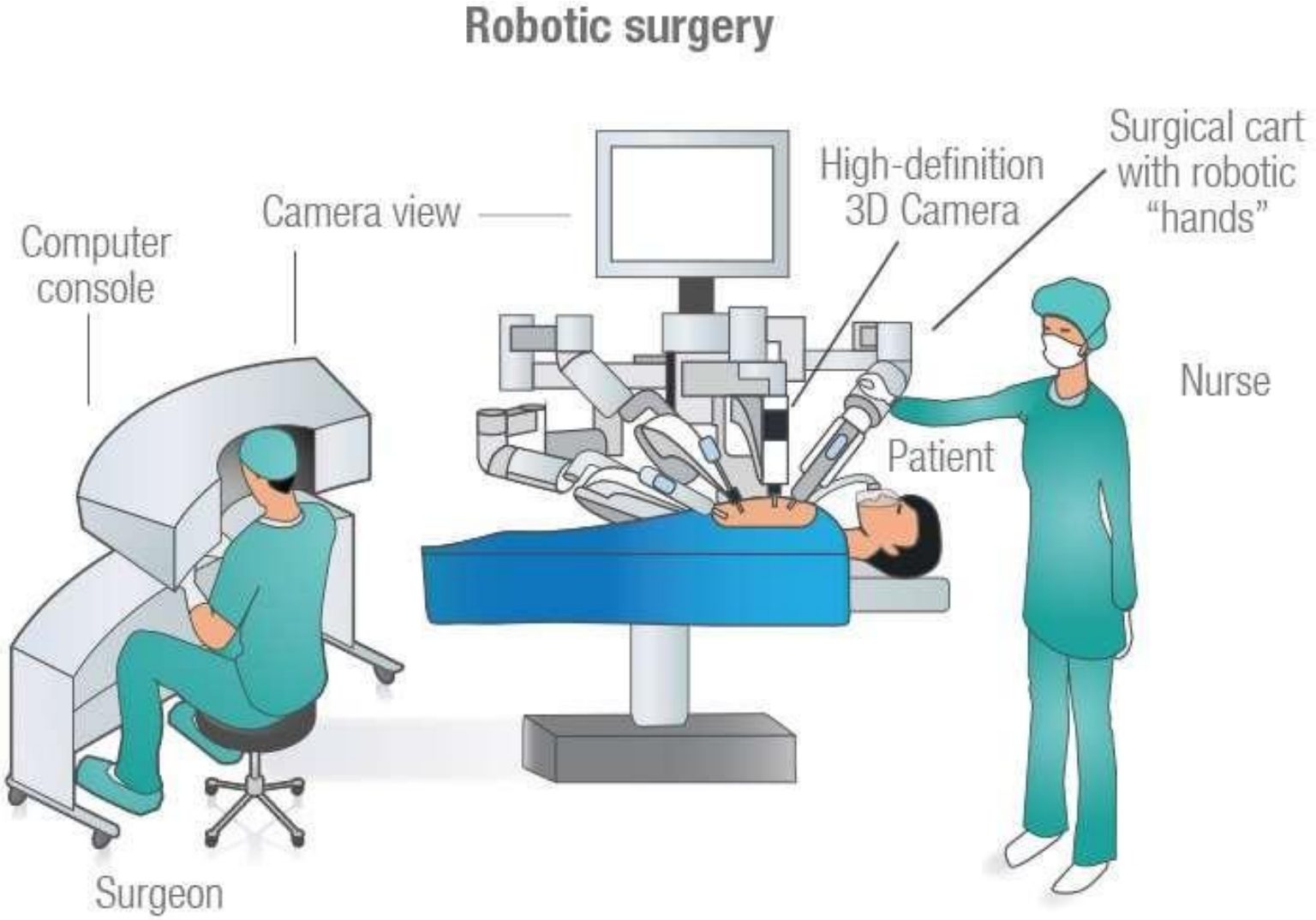
Future robots - Professional Robots in the field of Drug Delivery







Future robots - Surgery



Future Robots

- **Robotic** engineers are designing the next generation of **robots** to look, feel and act more human, to make it easier for us to warm up to a cold machine.
- Realistic looking hair and skin with embedded sensors will allow **robots** to react naturally in their environment.

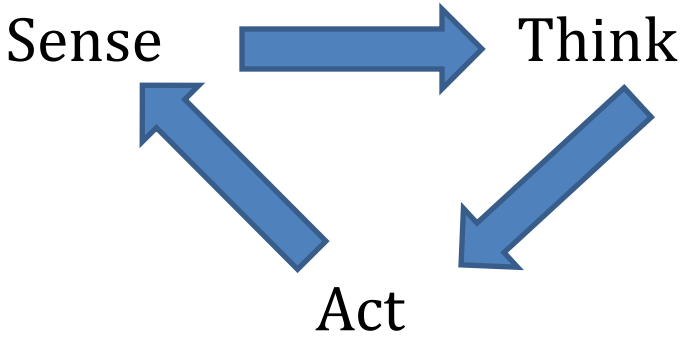
Robotics Terminology

- **Robot** - Mechanical device that performs human tasks, either automatically or by remote control.
- **Robotics** - Study and application of robot technology.
- **Telerobotics** - Robot that is operated remotely.

Robot Control Loop

Speech, Vision Acceleration, Temperature Position ,Distance Touch, Force Magnetic field ,Light Sound , PositionSense

Task planning Plan Classification Learn Process data Path planning Motion planning



Output information Move, Speech Text, Visuals Wheels Legs Arms Tracks

How Do Robots Work

A robot is any moving machine that can be programmed to perform tasks and gather information from its surroundings. Robots work from a central microprocessor that controls their movements, they also have sensors for examining the environment and power sources.

Classification of robots

Entire robots can be classified in to 2 broad categories:

Based on Applications

- Industrial Robots
- Tele Robots
- Explorer Robots
- Laboratory Robots
- Hobbyist Robots
- Educational Robots
- Medical Robots

Based on Configuration

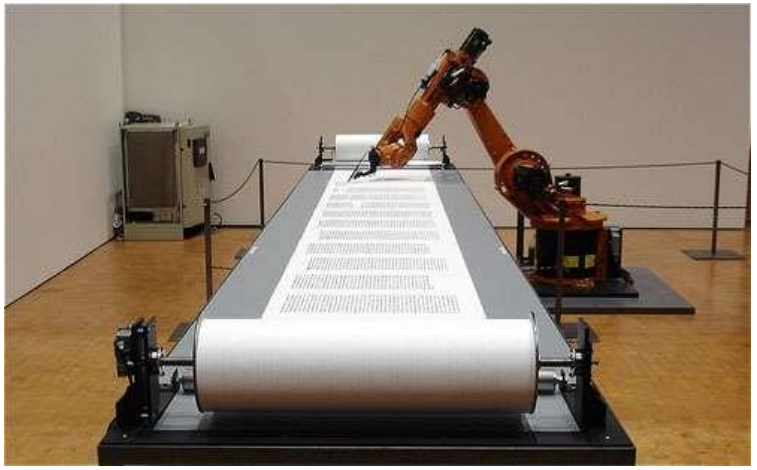
- Polar Configuration
- Cylindrical Configuration
- Cartesian Coordinate Configuration
- Jointed Arm Configuration

Based on Applications



Industrial Robots

- Materials handling
- Welding
- Inspection
- Improving productivity
- Laboratory applications



Classification Based on Applications

- Tele Robots



Classification Based on Applications

- Explorer Robots

They are used to go where humans cannot go and fear to go. Eg: to explore cave, in deep under water, to rescue people in sunken ships. Used in Hazardous Environments. Eg: Military application





Classification Based on Applications

- **Hobbyist Robots**

This category of robots are generally used for entertainment purpose and experimenting purpose. These robots usually equipped with speech synthesis techniques

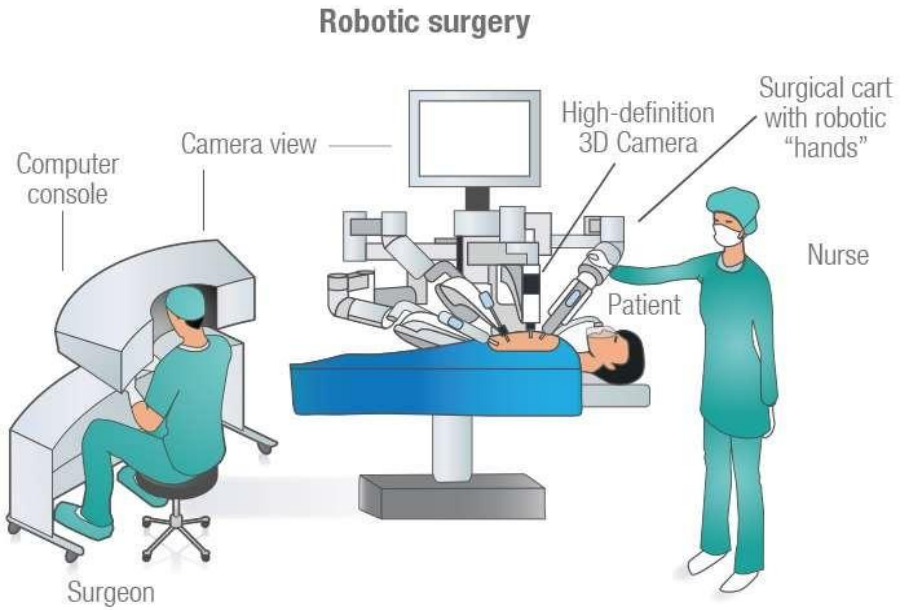




Classification Based on Applications

- **Medical Robots**

A **medical robot** is a robot used in the medical sciences. They include surgical robots. These are in most tele manipulators, which use the surgeon's actions on one side to control the "effector" on the other side.





- **Mobile Robots-**

–Robots that move around on legs, tracks or wheels.

Eg-

In 1979 a nuclear accident in the USA caused a leak of radioactive material which led to Production of special robot –which Can handle the radioactive materials.



•**Educational Robots** – Robotic kits Are used extensively in education. Eg-Robolab , Lego and RoboCup Soccer

•**Domestic Robots** those designed to perform household tasks and modern toys which are programmed to do things like talking, walking and dancing , etc.

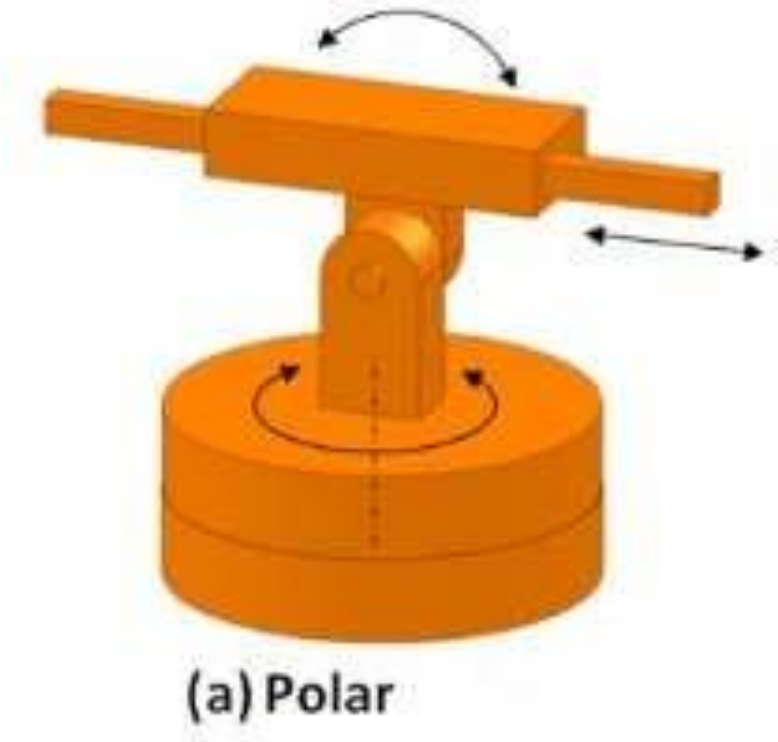




Classification Based on Configuration

- Polar Configuration

Has **one linear motion and two rotary motions**. It uses a telescoping arm that can be raised or lowered about a horizontal pivot.(α)
.The pivot is mounted on a rotating base.(θ). A arm also has the capability to move in and out to provide a linear motion(x). The various joints provide the robot with the capability to move its arm within a spherical space and hence it is also referred as Spherical coordinate robot.

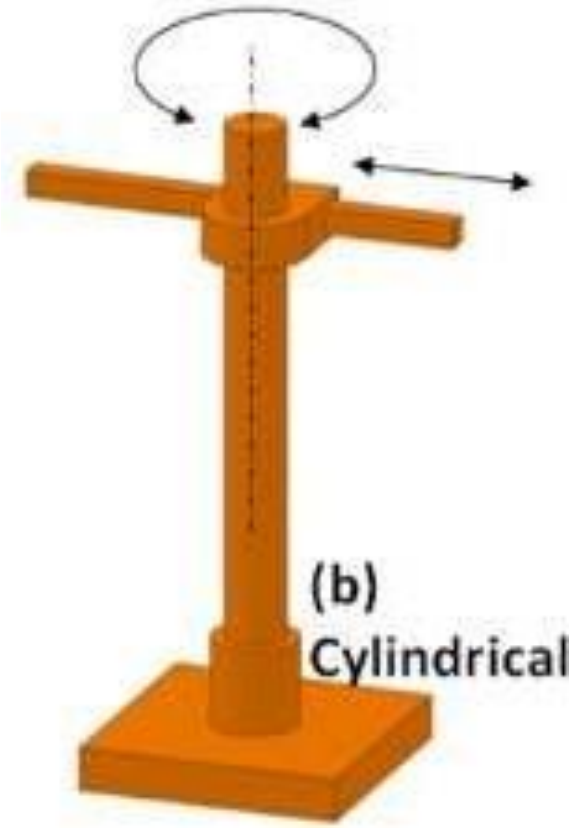


Classification Based on Configuration

Cylindrical Configuration

These robots uses a **vertical column and a slide** that can be moved up or down along the column. The robot arm is attached to the slide so that it can be moved radially with respect to the column. By rotating the column, the robot is capable of achieving a work space that approximates a cylinder.

Work Envelope: ***Cylinder***

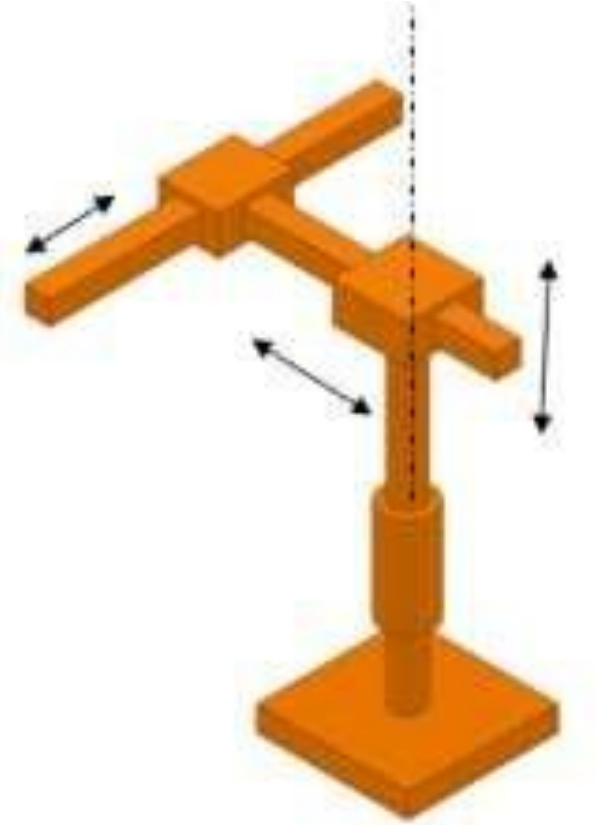


Classification Based on Configuration

Cartesian Coordinate Configuration

These robots uses **three perpendicular slides** to construct the *x,y* and *z* axes. Other names are sometimes applied to this configuration such as XYZ robots and rectilinear robot. By moving the three slides relative to one another, the robot is capable of operating within a rectangular work envelope.

Work Envelope: **Rectangular**



(c) Cartesian

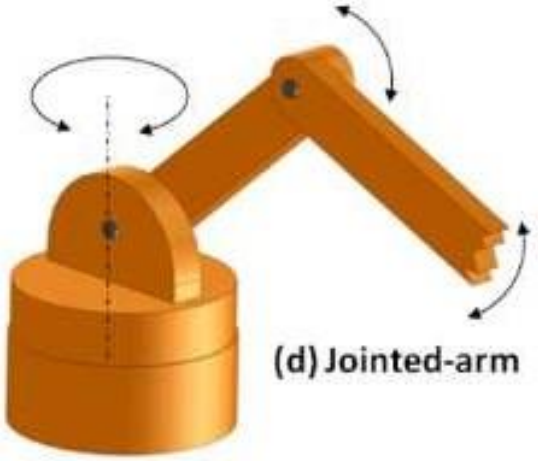
Classification Based on Configuration

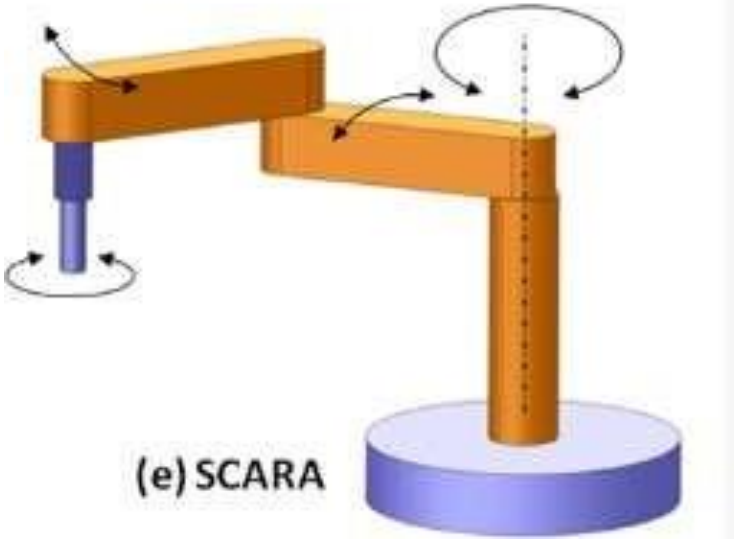
Jointed Arm Configuration

Jointed arm configuration can be classified in to 2 types:

(i)Jointed Arm Vertical Configuration

(ii)Jointed Arm Horizontal Configuration (SCARA - Selective Compliance Articulated Robot Arm)





Laws of Robotics

- Asimov proposed three “Laws of Robotics”
- Law 1: A robot may not injure a human being or through inaction, allow a human being to come to harm.
- Law 2: A robot must obey orders given to it by human beings except where such orders would conflict with the first law.
- Law 3: A robot must protect its own existence as long as such protection does not conflict with the first law and Second law.

Robot Components

1. **Manipulator or Rover**: Main body of robot (Links, Joints, other structural element of the robot)
2. **End Effecter**: The part that is connected to the last joint hand) of a manipulator.
3. **Actuators**: Muscles of the manipulators (servomotor, stepper motor, pneumatic and hydraulic cylinder).
4. **Sensors**: To collect information about the internal state of the robot or To communicate with the outside environment.

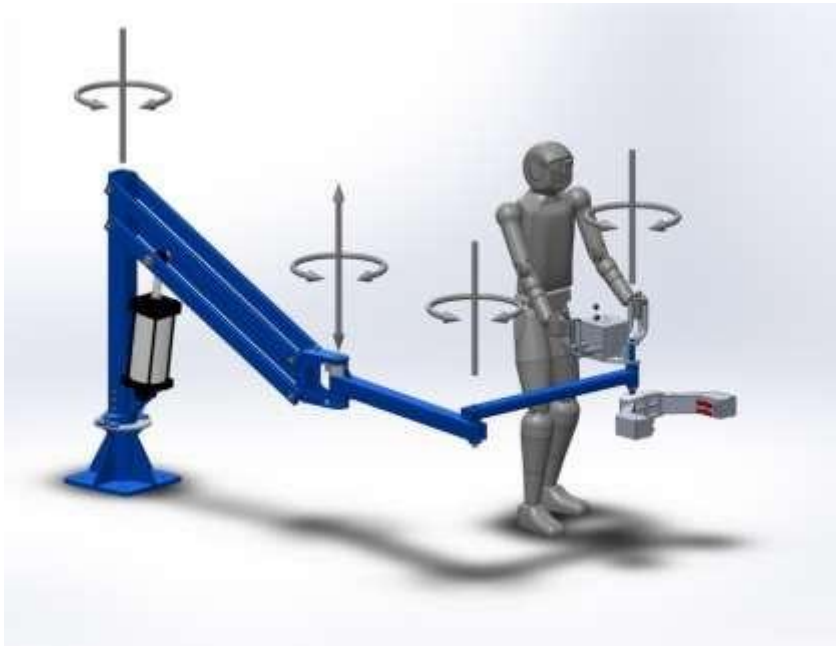
5. **Controller**: Similar to cerebellum. It controls and coordinates the motion of the actuators.
6. **Processor**: The brain of the robot. It calculates the motions and the velocity of the robot's joints, etc.
7. **Software**: Operating system, robotic software and the collection of routines.

Basic Parts of a Robot

- **Manipulator / Rover** :

This is the main body of the Robot and consists of links, joints and structural elements of the Robot.





Basic Parts of a Robot

- **End Effectors** : This is the part that generally handles objects, makes connection to other machines, or performs the required tasks. It can vary in size and complexity according to the applications / requirements.







End Effectors

- In robotics, an end effectors is the device at the end of a robotic arm, designed to interact with the environment.

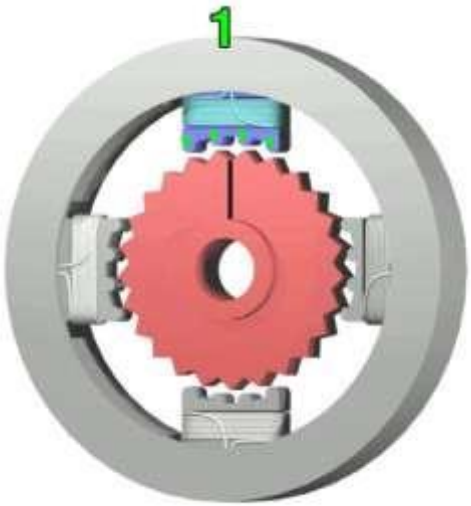
- End effectors may consist of a gripper or a tool. The gripper can be of two fingers, three fingers or even five fingers.



Basic Parts of a Robot

- **Actuators** : Actuators are the muscles of the manipulators. Common types of actuators are servomotors, stepper motors, pneumatic cylinders etc.







Actuators

- Locomotion
- Manipulation

- **Locomotion-**
- Legs
- Wheels
- Other exotic means







Basic Parts of a Robot

- **Sensors :** Sensors are used to collect information about the internal state of the robot or to communicate with the outside environment. Robots are often equipped with external sensory devices such as a vision system, touch and tactile sensors etc which help to communicate with the environment



Touch Sensor



Ultrasonic Sensor



Color Sensor



Sound Sensor



Light Sensor

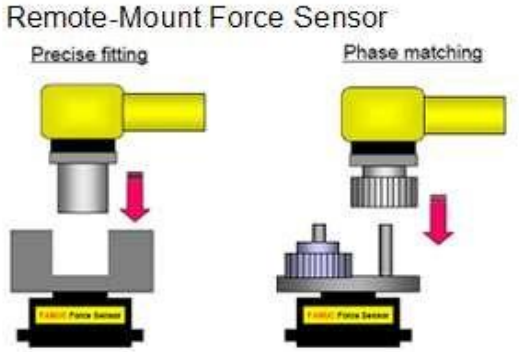


Motor Sensor

Sensors

- Sensors provide awareness of the environment by sensing things. Sensors are the core of robots. It is the system that alerts the robots.
- Sensing can be in different forms like-

- Light
- Sound
- Heat
- Chemicals
- Force
- Object proximity
- Physical orientation/position
- Magnetic & Electric Fields
- Resistance







Touch Sensor



Basic Parts of a Robot

- **Controller :** The controller receives data from the computer, controls the motions of the actuator and coordinates these motions with the sensory





Robot Anatomy

ROBOT ANATOMY AND RELATED ATTRIBUTES

- The anatomy of industrial robots deals with the assembling of outer components of a robot such as wrist, arm and body.
- Before jumping into robot configurations, here are some of the key facts about robot anatomy.

(a) Joints and Links

(b) Common Robot Configurations

JOINTS AND LINKS

- The manipulator of an industrial robot consists of a series of joints and links.
- Robot anatomy deals with the study of different joints and links and other aspects of the manipulator's physical construction.
- A robotic joint provides relative motion between two links of the robot.
- Each joint, or axis, provides a certain degree-of-freedom (DOF) of motion.
- In most of the cases, only one degree-of-freedom is associated with each joint.
- Robot's complexity can be classified according to the total number of degrees -of-freedom they possess.
- Each joint is connected to two links, an input link and an output link.

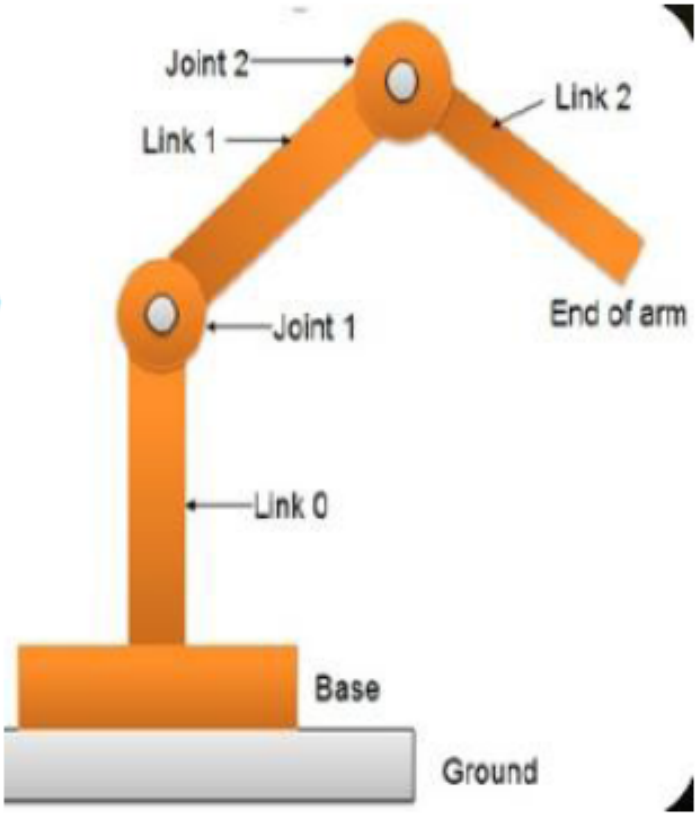
JOINTS AND LINKS

- A Joint provides controlled relative movement between the input link and output link. A robotic link is the rigid component of the robot manipulator.

- Most of the robots are mounted upon a stationary base, such as the floor. From this base, a joint-link numbering scheme may be recognized as shown in
Figure.

JOINTS AND LINKS

- The robotic base and its connection to the first joint are termed as link-0.
- The first joint in the sequence is joint-1.
- Link-0 is the input link for joint-1, while the output link from joint-1 is link-1 which leads to joint-2.
- Link 1 is the output link for joint-1 and the input link for joint-2.
- This joint-link-numbering scheme is further followed for all joints and links in the robotic systems.

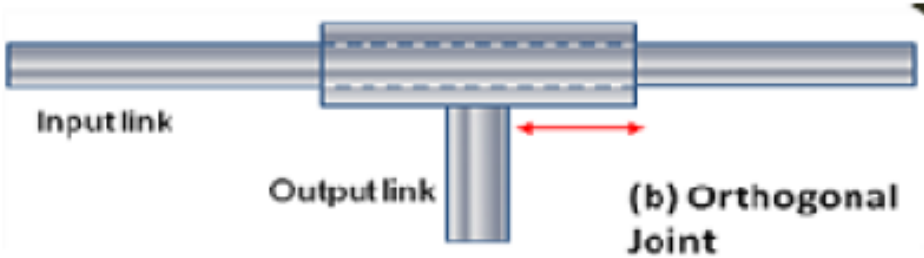


JOINTS AND LINKS

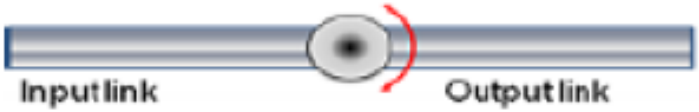
• Nearly all industrial robots have mechanical joints that can be classified into following five types as shown in Figure below.



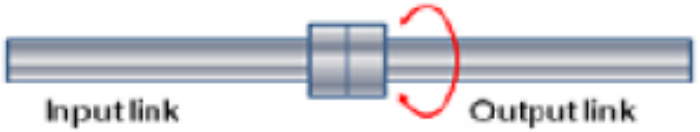
(a) Linear Joint



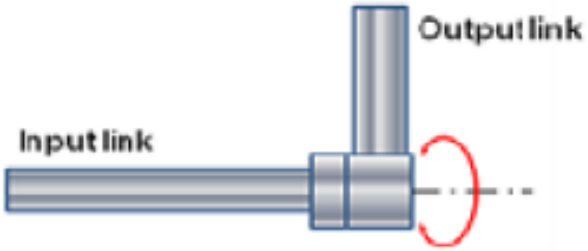
(b) Orthogonal Joint



(c) Rotational Joint



(d) Twisting Joint



(e) Revolving Joint

JOINTS AND LINKS

a) Linear joint (type L joint)

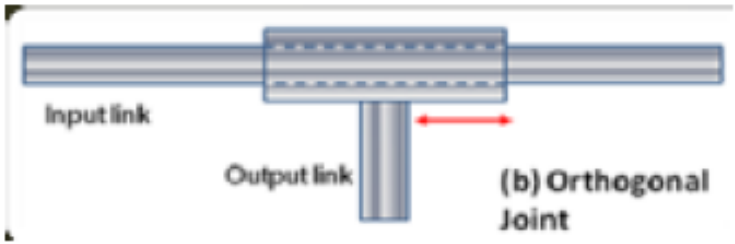
• The relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links being parallel.



(a) Linear Joint

b) Orthogonal joint (type U joint)

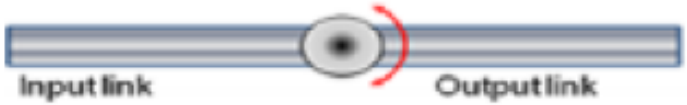
• This also has a translational sliding motion, but the input and output links are perpendicular to each other during the move.



(b) Orthogonal Joint

c) Rotational joint (type R joint)

• This type provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links. 7



(c) Rotational Joint

JOINTS AND LINKS

d) Twisting joint (type T joint)

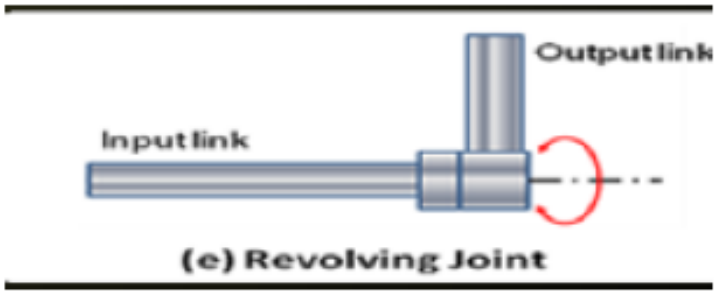
- This joint also involves rotary motion, but the axis or rotation is parallel to the axes of the two links.



(d) Twisting Joint

e) Revolving joint (type V-joint, V from the "v" in revolving)

- In this type, axis of input link is parallel to the axis of rotation of the joint. Axis of the output link is perpendicular to the axis of rotation.

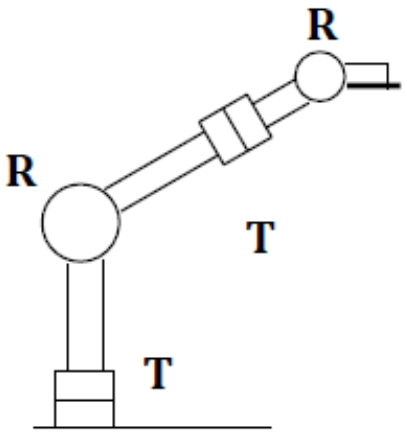


(e) Revolving Joint

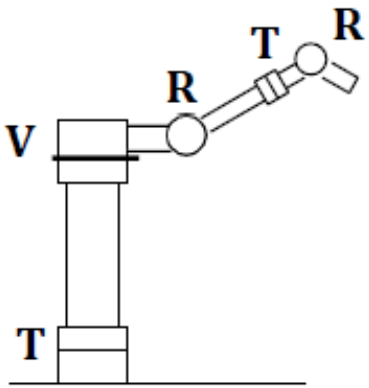
EXAMPLE

- Sketch following manipulator configurations
- (a) TRT:R, (b) TVR:TR, (c) RR:T.

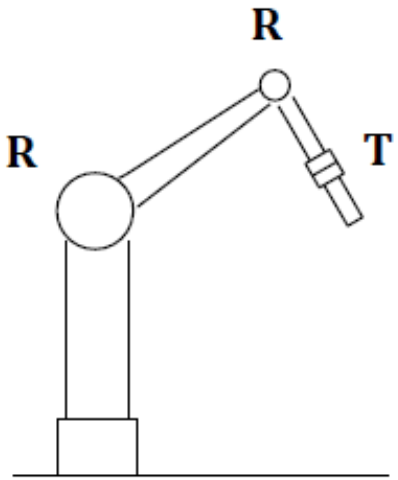
Solution:



(a) TRT:R

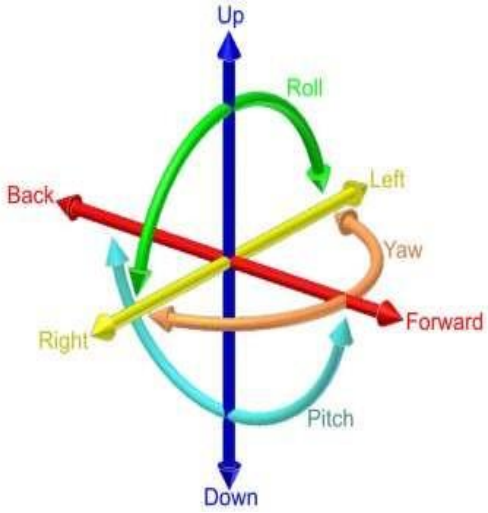


(b) TVR:TR



(c) RR:T

DEGREES OF FREEDOM



- The ***degrees of freedom*** (dof) of a rigid body is defined as the number of independent movements it has.
- Higher number dof indicates an increased flexibility in positioning a tool.
- For each degree of freedom a joint is required.
- The degrees of freedom located in the arm define the configuration.
- Three degrees of freedom located in the wrist give the end effector all the flexibility.
- A total of six degrees of freedom is needed to locate a robot's hand at any point in its work space.

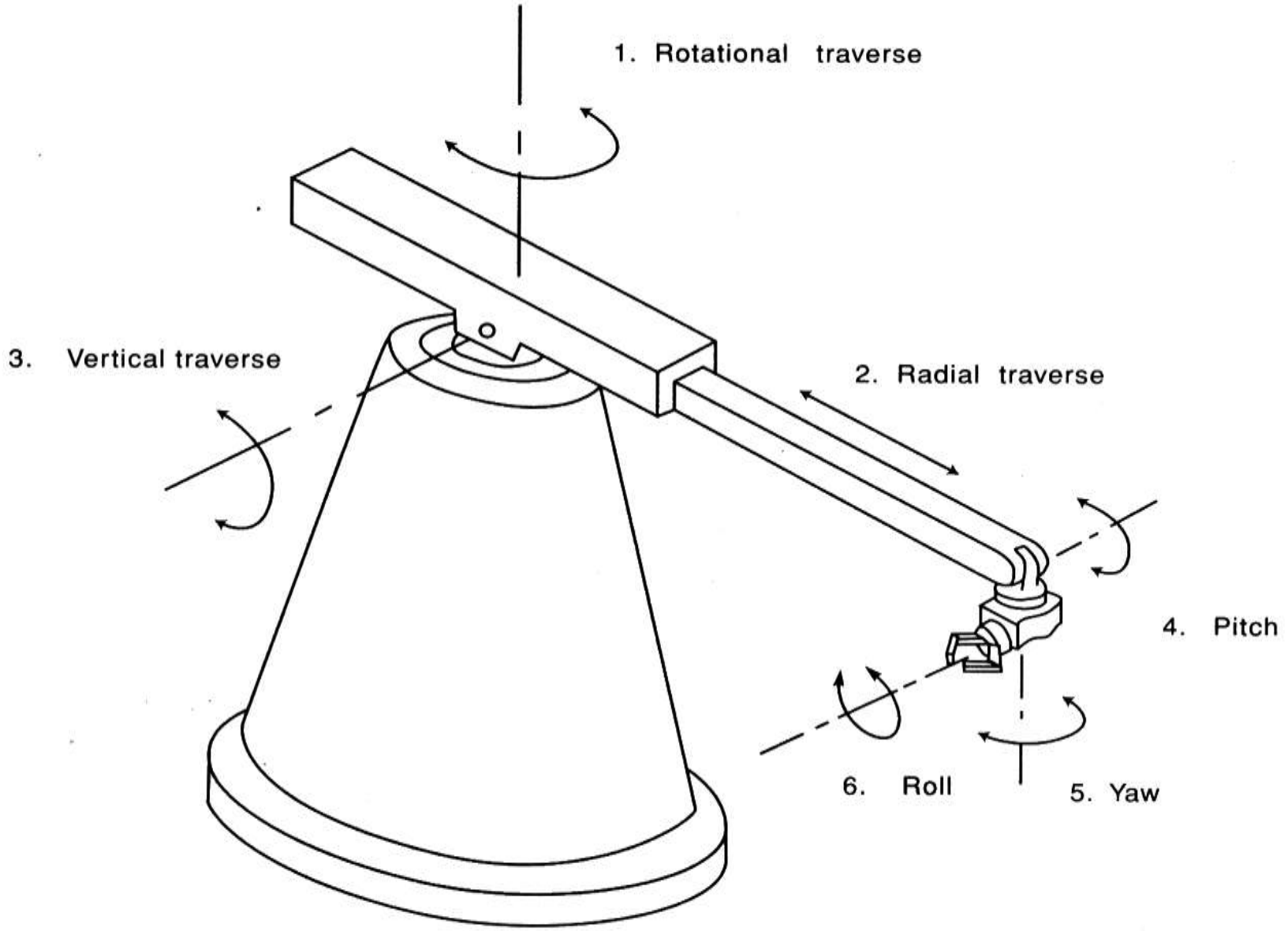


Figure 3.3.2 Six major degrees of freedom of a robotic system

ROBOT ARM

- The three degrees of freedom located in the arm of a robotic system are:
  - The rotational traverse: is the movement of the arm assembly about a rotary axis, such as left-and-right swivel of the robot's arm about a base.
  - The radial traverse: is the extension and retraction of the arm or the in-and-out motion relative to the base.
  - The vertical traverse: provides the up-and-down motion of the arm of the robotic system.

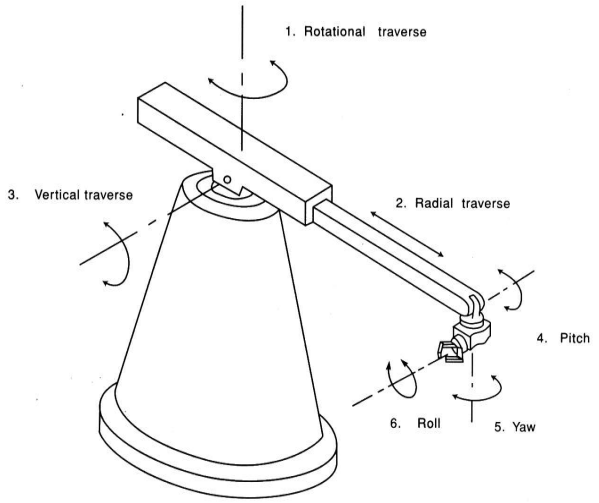
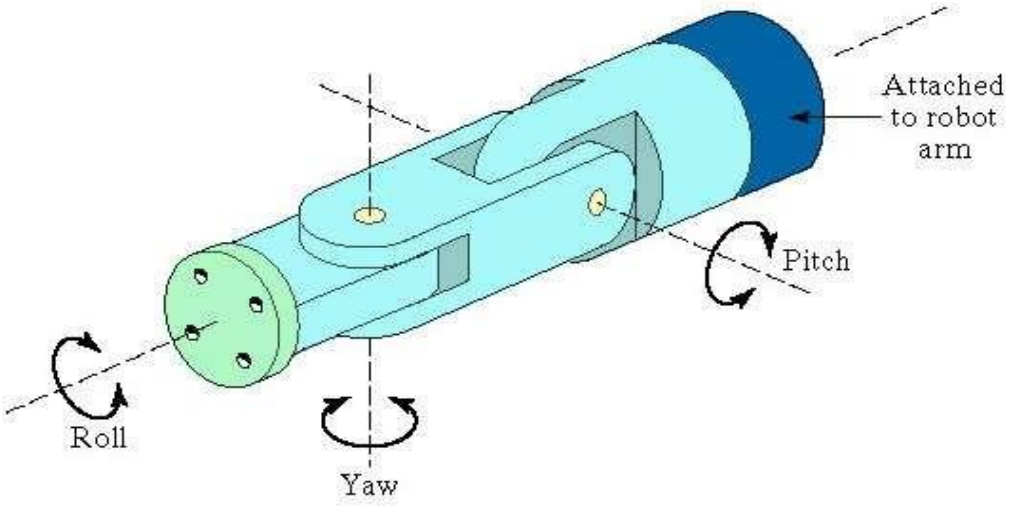


Figure 3.3.2 Six major degrees of freedom of a robotic system

ROBOT WRIST

- Wrist assembly is attached to end-of-arm
- End effector is attached to wrist assembly
- Function of wrist assembly is to orient end effector
  - Body-and-arm determines global position of end effector
- Two or three degrees of freedom:
  - Roll
  - Pitch
  - Yaw



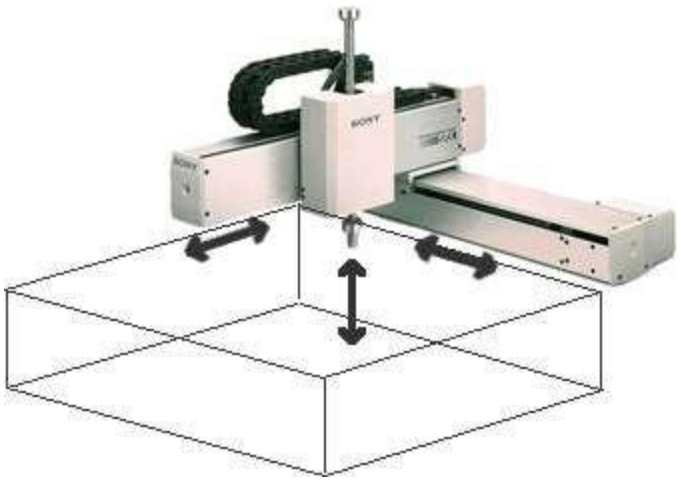
- The three degrees of freedom located in the wrist, which bear the names of aeronautical terms, are
  - Pitch or bend: is the up-and-down movement of the wrist.
  - Yaw: is the right-and-left movement of the wrist.
  - Roll or swivel: is the rotation of the hand.

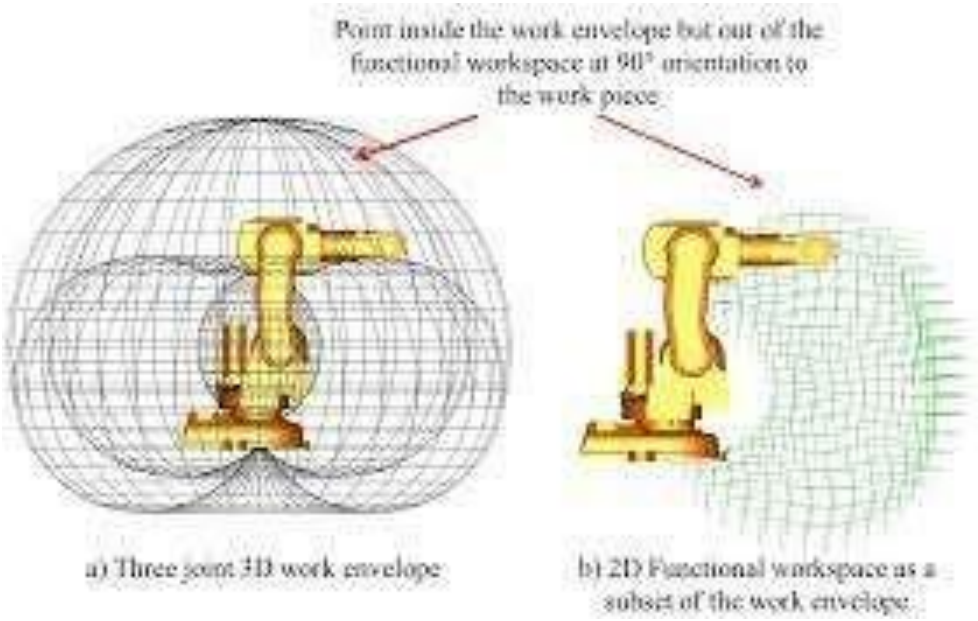
Defining Parameters for industrial robots

- **Degrees of freedom–** number of independent motions that are allowed to the body, this is usually the same as the number of axes.

Defining Parameters for industrial robots

- **Working envelope–** an **envelope** is the region of space a robot can reach during its normal range of motion.

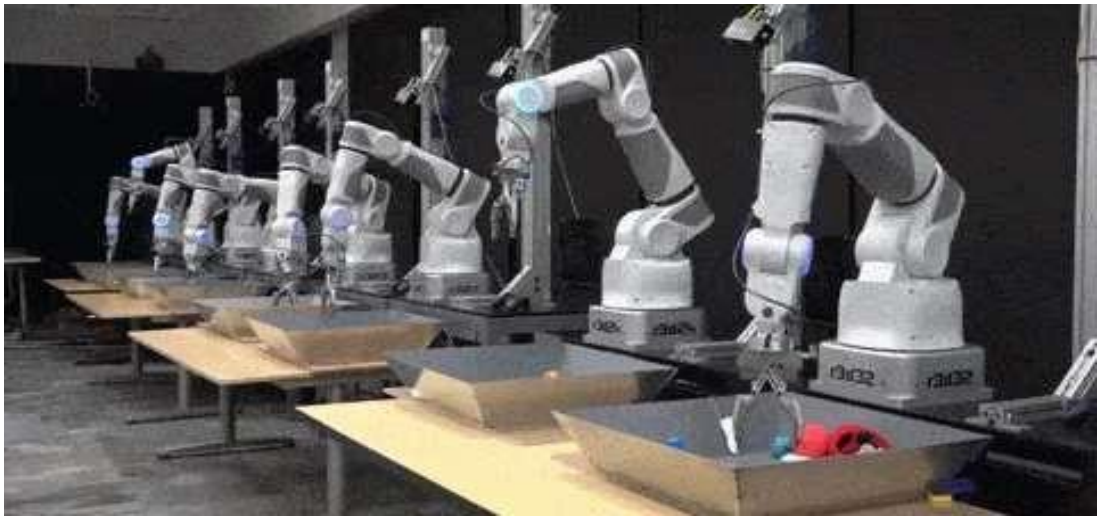


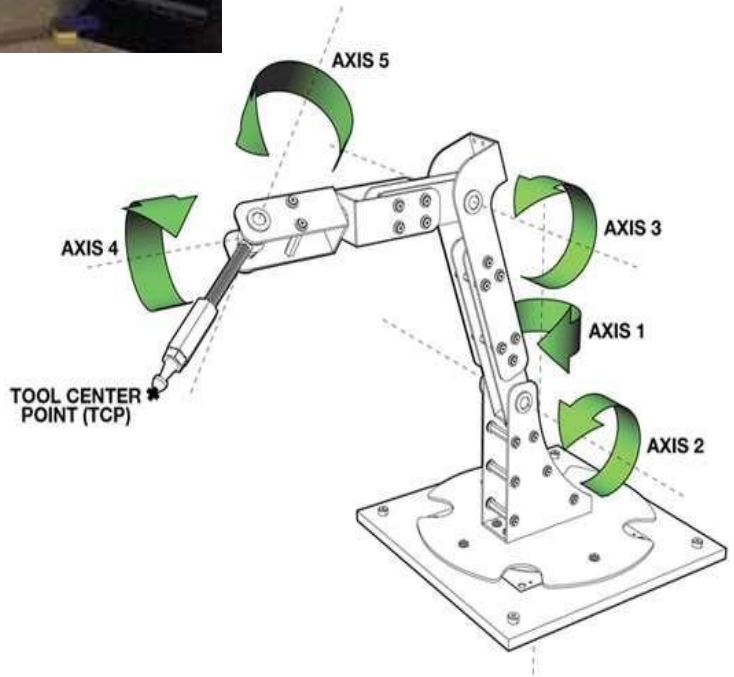


Defining Parameters for industrial robots

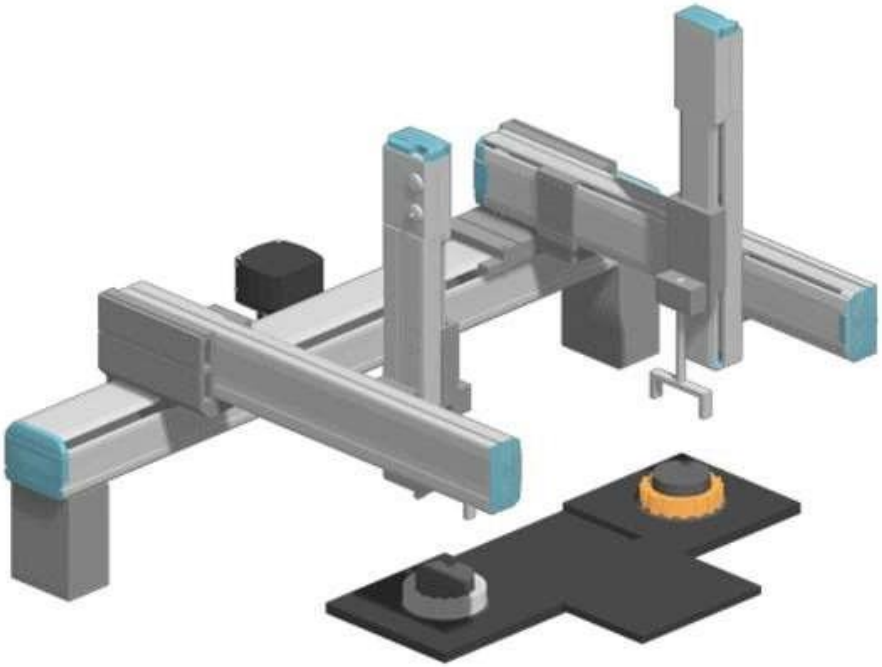
- **Kinematics** – the actual arrangement of rigid members and joints in the robot, which determines the robot's possible motions.
- Classes of robot kinematics include articulated, cartesian, parallel and SCARA.

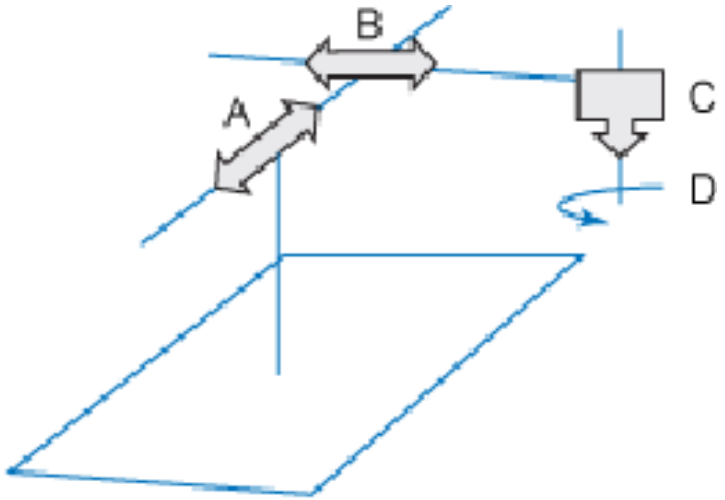
Kinematics – articulated robot





Kinematics – cartesian robots

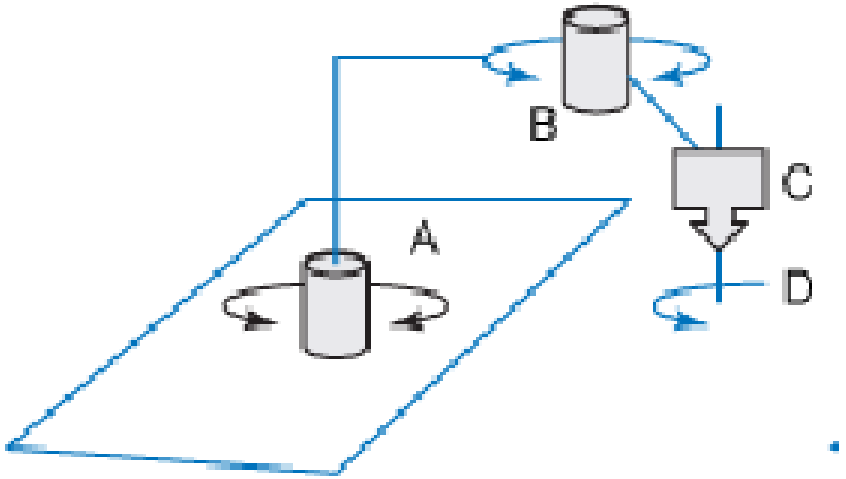




CARTESIAN 4-DOF

Kinematics – scara robots





SCARA 4-DOF

Defining Parameters for industrial robots

- **Carrying capacity or payload** – how much weight a robot can lift.
- **Speed**– how fast the robot can position the end of its arm. This may be defined in terms of the angular or linear speed of each axis or as a compound speed i.e. the speed of the end of the arm when all axes are moving.
- **Acceleration** – how quickly an axis can accelerate. Since this is a limiting factor a robot may not be able to reach its specified maximum speed for movements over a short distance or a complex path requiring frequent changes of direction.

Defining Parameters for industrial robots

- ***Accuracy*** – how closely a robot can reach a commanded position. When the absolute position of the robot is measured and compared to the commanded position the error is a measure of accuracy. Accuracy can be improved with external sensing, for example a vision system or Infra-Red. Accuracy can vary with speed and position within the working envelope and with payload.

- ***Repeatability*** – how well the robot will return to a programmed position. This is not the same as accuracy. It may be that when told to go to a certain X-Y-Z position that it gets only to within 1 mm of that position. This would be its accuracy which may be improved by calibration. But if that position is taught into controller memory and each time it is sent there it returns to within 0.1mm of the taught position then the repeatability will be within 0.1mm.

Accuracy vs Repeatability



The Purpose of Robots

- Robots are also used for the following tasks:

- Dirty Tasks
- Repetitive tasks
- Dangerous tasks
- Impossible tasks
- Robots assisting the handicapped
- Can operate equipments at much higher precision than humans.

Robotics Application

- **EXPLORATION-**
  - Space Missions
  - Robots in the Antarctic
  - Exploring Volcanoes
  - Underwater Exploration
- **MEDICAL SCIENCE**
  - Surgical assistant
- **ASSEMBLY**- factories Parts-
  - Handling
  - Assembly
  - Painting
  - Surveillance
  - Security (bomb disposal , etc)
  - Home help (grass cutting, nursing)





Advantages

- Going to far away planets.
- Going far down into the unknown waters and mines where humans would be crushed
- Giving us information that humans can't get
- Working at places 24/7 without any salary and food. Plus they don't get bored
- They can perform tasks faster than humans and much more consistently and accurately
- Most of them are automatic so they can go around by themselves without any human interference.

Disadvantages

- People can lose jobs in factories
- It needs a supply of power
- It needs maintenance to keep it running .
- It costs money to make or buy a robot

Future Prospect

- Scientists say that it is possible that a robot brain will exist by 2019 .
- **Vernor Vinge** has suggested that a moment may come when computers and robots are smarter than humans.
- In 2009, some robots acquired various forms of semi-autonomy, including being able to find power sources on their own.